

United International University (UIU)

Department of Computer Science and Engineering

A Survey on Adversarial Patch Attacks on Target Detection

Course Project Report

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1 Introduction

Object detection is one of the key topics of computer vision because of its application in various areas, including autonomous driving, surveillance, robotics, healthcare, and remote sensing. One of the most advanced object detection algorithms is represented by the family of You Only Look Once (YOLO) algorithms, which became popular owing to their high object detection accuracy and real-time performance. YOLO models have been evolving with the help of improvements in network architectures, feature fusion, detection heads, and training procedures to improve their accuracy, speed, and efficiency.

Notwithstanding excellent performance of the latest YOLO models, they can be easily attacked with the help of carefully created adversarial perturbations that can trick object detectors, being almost undetectable for humans. Therefore, nowadays increasing attention is being paid not only to the accuracy of the object detection model but also to the robustness of such models to adversarial perturbations. At the same time, Diffusion models become more and more popular as an alternative to traditional approaches to creating adversarial samples because of the possibility to create realistic images with high success rates of attack.

This literature review examines eight papers (from 2025-2026), with four on YOLO-based models and four on Diffusion-based attacks. The results demonstrate recent trends in object detection, robustness to adversarial attacks, and the generation of Diffusion-based attacks, pointing towards the necessity for robust and intelligent detection systems.

Literature Review

2 Comparative Analysis of YOLO-Based Studies

Table 1: Comparative Analysis of YOLO-Based Studies

Paper	Abstract /Paper Summary	Introduction	Dataset	Methodology	Future Direction / Limitations
Paper 1 *YOLOv1 to YOLOv11: A Comprehensive Survey of Real-Time Object Detection Innovations and Challenges*	Reviews the evolution of YOLOv1–YOLOv11, highlighting architectural improvements, benchmark performance, and real-world applications.	Summarizes the architectural evolution of YOLO models, compares benchmark performance, and discusses deployment challenges and future research opportunities.	MS COCO, Pascal VOC, ImageNet	Systematic survey comparing YOLO versions based on backbone, neck, detection head, training strategy, mAP, and FPS.	Future work focuses on robustness, lightweight deployment, small-object detection, and model interpretability.
Paper 2 *Hyper-YOLO: When Visual Object Detection Meets Hypergraph Computation*	Introduces Hyper-YOLO by incorporating hypergraph computation to improve feature representation and multi-scale object detection.	Addresses limitations of conventional feature fusion and proposes hypergraph learning for richer semantic information.	MS COCO	Uses MANet, HyperC2Net, and HGC-SCS for hypergraph-based feature extraction and multi-scale feature fusion.	Computational complexity should be further reduced while preserving high detection accuracy for real-time applications.
Paper 3 *Delving into YOLO Object Detection Models: Insights into Adversarial Robustness*	Evaluates the robustness of modern YOLO models against adversarial attacks and image corruption.	Investigates the reliability of YOLO models under challenging conditions and analyzes their robustness.	MS COCO	Compares YOLO models using FGSM, PGD, image corruption, IoU, AP, and mAP.	Future work should improve adversarial robustness and detection reliability in safety-critical environments.
Paper 4 *YOLO Advances to Its Genesis: A Decadal and Comprehensive Review of the YOLO Series*	Reviews the decade-long development of the YOLO family from YOLOv1 to YOLOv12.	Discusses technological evolution, architectural innovations, and emerging research trends in YOLO.	MS COCO, Pascal VOC, ImageNet	Chronological review comparing architectures, benchmark performance, efficiency, and application domains.	Future research should emphasize multimodal learning, lightweight models, efficient deployment, and intelligent object detection.

Discussion

From the comparative analysis, it has been identified that the research related to YOLO has seen progression through three distinct stages. The first stage involved development of methods that could improve real-time object detection capabilities through improvements in network design and training approaches (Papers 1 and 2). In the second stage, there was greater emphasis on the robustness of models from adversarial attacks and corruption attacks, underlining the need for robustness in safety-sensitive applications (Paper 3). The latest stage involves the use of sophisticated feature learning methods including hypergraph computation that can enhance the capabilities of models in multi-scale feature representations (Paper 4).

2.1 Performance Comparison Analysis of YOLO-Based Studies

In order to understand how far the field of YOLO-based object detection has come, a comparison is made of the four selected papers in terms of objectives, methods, data sets, and major conclusions drawn by each study. While Papers 1 and 2 offer detailed surveys of the development history of the YOLO series of algorithms, Papers 3 and 4 aim at improving the robustness and performance of the algorithm using experimentation and architecture design.

Table 2: Performance Comparison Analysis of YOLO-Based Studies

Paper	**Objective**	**Dataset**	**Evaluation Metrics**	**Key Findings**
Paper 1: YOLOv1 to YOLOv11: A Comprehensive Survey	Reviews the evolution of YOLO models from YOLOv1 to YOLOv11	MS COCO, Pascal VOC, ImageNet	mAP, FPS, Architecture Comparison	Consecutive developments in the YOLO architecture contribute greatly to better accuracy, processing speed, and efficiency..
Paper 2: YOLO Advances to Its Genesis	Reviews the decade-long development of the YOLO series	MS COCO, Pascal VOC, ImageNet	mAP, Latency, Computational Efficiency	The modern version of YOLO provides higher accuracy and lower latency, thus being applicable for real-time systems.
Paper 3: Delving into YOLO Object Detection Models: Insights into Adversarial Robustness	Evaluates robustness under adversarial attacks and image corruption	MS COCO	mAP, IoU, AP, Robustness Analysis	YOLOX (large) demonstrates enhanced robustness to adversarial examples, but all YOLO-based models exhibit degraded performance when subjected to heavy perturbations.
Paper 4: Hyper-YOLO: When Visual Object Detection Meets Hypergraph Computation	Proposes Hyper-YOLO using hypergraph computation	MS COCO	AP, AP50, APs, APm, APl, FPS	Hyper-YOLO consistently outperforms YOLOv8, YOLOv9, and Gold-YOLO, achieving 41.8% AP (-N), 48.0% AP (-S), 52.0% AP (-M), and 53.8% AP (-L) while maintaining efficient inference speed.

Performance Comparison Discussion

Table 2 compares the results of the four latest works based on YOLO framework. Works 1 and 2 emphasize the continuous development of the YOLO framework in terms of detection precision, inference speed, and efficiency by the way of architectural advances. Work 3 shows that even though new YOLO architectures are robust to the adversarial attacks, their performances still drop under adversarial perturbations. Meanwhile, Work

4 presents the highest performance in detection using the Hyper-YOLO approach that introduces hypergraph-based features learning and boosts Average Precision (AP) performance. In general, the table reflects the fact that modern YOLO works aim at finding a trade-off between detection precision, robustness, and inference speed.

3 Comparative Analysis of Diffusion-Based Studies

Table 3: Comparative Analysis of Diffusion-Based Studies

Paper	Abstract / Paper Summary	Introduction	Dataset	Methodology	Future Direction / Limitations
Paper 5 *ObjectAdv Object-Level Adversarial Attacks (AAAI 2026)*	Proposes **ObjectAdv**, a diffusion-based object-level unrestricted adversarial attack framework that generates realistic adversarial examples by perturbing only the target object.	Introduces an object-level diffusion framework to overcome the trade-off between attack effectiveness and image quality in unrestricted adversarial attacks.	ImageNet-Compatible Dataset (1,000 ImageNet validation images)	Utilizes **Stable Diffusion**, attention-based object localization, prompt-switching, FFT-based edge smoothing, and multi-loss optimization to generate natural adversarial examples.	Future work should improve attack transferability, computational efficiency, and robustness against advanced defense mechanisms.
Paper 6 *DiffProtect Facial Privacy*	Proposes **DiffProtect**, a diffusion-based framework for generating natural adversarial face images to protect facial privacy.	Uses diffusion models to generate semantic adversarial examples while preserving facial appearance and improving attack success against face recognition systems.	CelebA-HQ, FFHQ	Employs **Diffusion Autoencoder (DiffAE)**, conditional DDIM decoder, semantic regularization, and DiffProtect-fast for efficient adversarial image generation.	The iterative diffusion process increases inference time; future work should improve efficiency while maintaining image quality and attack performance.
Paper 7 *DiffAdvMAP (ICML 2025)*	Introduces **DiffAdvMAP**, a diffusion-based framework that generates natural unrestricted adversarial examples using MAP optimization.	Addresses the balance between attack effectiveness, transferability, and image realism by integrating diffusion models with posterior optimization.	ImageNet	Combines diffusion models, MAP optimization, reconstruction constraints, adversarial constraints, and destruction–reconstruction strategies to improve unrestricted adversarial attacks.	Future work should enhance black-box transferability, inference efficiency, and robustness against stronger defense techniques.
Paper 8 *YOLO, Transformer & Diffusion (2025)*	Reviews YOLO, Transformer, and diffusion-based methods for small object detection and compares their strengths and weaknesses.	Provides a unified overview of three major detection paradigms and identifies challenges in achieving accurate and efficient small object detection.	MS COCO, PASCAL VOC, Crowd-Human, VisDrone, COCO-DE, VOC-DE	Reviews **DiffusionDet**, **DiffCOD**, and **DiffusionEngine** for bounding-box generation, feature fusion, and diffusion-based data generation.	Future research should develop lightweight diffusion architectures, improve multimodal feature fusion, reduce computational cost, and support real-time deployment.

Discussion

The comparative study reveals the fast development of diffusion-based approaches for various computer vision problems. The first paper (Paper 5) describes unrestricted adversarial attacks at an object level that provide high attack success rates and good image

quality at the same time. The second paper (Paper 6) introduces diffusion models for protecting people's privacy by generating real-life-like adversarial faces. Paper 7 increases the power of unrestricted adversarial attacks using MAP optimization with diffusion models to increase transferability and robustness. Finally, Paper 8 provides a review on diffusion-based approaches for small object detection where better detection accuracy is reported while computation cost and inference speed are considered major disadvantages. Overall, it can be said that diffusion models became one of the most popular tools for generating adversarial attacks, protecting privacy, and detecting objects.

3.1 Performance Comparison of Selected Diffusion-Based Studies

Table 4: Performance Comparison of Selected Diffusion-Based Studies

Paper	Dataset	Evaluation Metrics	Performance / Results
Paper 5 *ObjectAdv Object-Level Adversarial Attacks (AAAI 2026)*	ImageNet-Compatible Dataset	Attack Success Rate (ASR), SSIM, FID	Achieved **99.2%** white-box ASR, SSIM of **0.914**, and FID of **25.63**, outperforming previous unrestricted adversarial attack methods while maintaining high visual quality.
Paper 6 *DiffProtect Facial Privacy*	CelebA-HQ, FFHQ	Attack Success Rate (ASR), Cosine Similarity, Image Quality	Achieved higher attack success rates than existing facial privacy protection methods while preserving natural facial appearance through semantic diffusion-based perturbations.
Paper 7 *DiffAdvMAP (ICML 2025)*	ImageNet	Attack Success Rate (ASR), Transferability, Image Quality	Achieved nearly **100%** white-box ASR, improved black-box transferability, and generated natural unrestricted adversarial examples with lower perceptual distortion than previous methods.
Paper 8 *YOLO, Transformer & Diffusion (2025)*	MS COCO, PASCAL VOC, CrowdHuman, VisDrone	AP, mAP, Precision, Recall (Survey)	Reported that diffusion-based object detectors generally provide higher detection accuracy for small objects than conventional detectors, although they require higher computational cost and longer inference time.

3.2 Performance Comparison Discussion

The Diffusion-based papers reviewed here exemplify the increasingly greater efficiency of Diffusion models not only in adversarial attacks but also in object detection tasks. Paper 5 is notable for the highest white-box attack efficiency (99.2%) while preserving good quality visually due to object-based perturbations. Paper 6 shows how Diffusion models can be applied to protect people's faces in the era of facial privacy violations in order to generate natural adversarial face images while improving attack efficiency and preserving semantic consistency. Paper 7 develops unrestricted adversarial attacks with Diffusion models even further by introducing maximum *a posteriori* (MAP) optimization in order to obtain nearly 100% attack efficiency and better black-box transferability. Lastly, Paper 8 highlights the benefits of Diffusion-based object detection methods especially in the case of small object detection tasks and achieves better detection accuracy

compared to traditional object detection algorithms while requiring higher computational costs and lower speeds. Overall, the above papers prove that the introduction of diffusion models allows for producing more realistic adversarial samples and improving detection capabilities. However, computational efficiency remains an issue.

4 Conclusion

A synthesis of eight papers of the latest advances (2025–2026) in the field of object detection via YOLO and diffusion-based attacks generation will be provided in this literature review. As is seen from the YOLO papers review, because of the continuous improvements in the sphere of feature fusion, hypergraph computation, and detection, considerable progress has been achieved in the area of accuracy, computational efficiency, and real-time object detection. It should be pointed out that, even taking into account all the advantages, latest YOLO models are not resistant to adversarial attacks.

From the review of diffusion models, it may be noted that diffusion models prove to be quite efficient when it comes to the generation of highly visual realistic adversarial examples with the preservation of high quality. The recent models like ObjectAdv, DiffProtect, and DiffAdvMAP produce highly successful attacks compared to previous methods. Besides, there exists a great potential of diffusion models in the development of small object detection due to their good feature generation and data augmentation capabilities. Unfortunately, the high computational cost and low inference speed constitute serious obstacles for the implementation of this method.

In conclusion, it may be stated that further researches in the sphere of object detection should be aimed at the development of efficient, accurate, light, and resistant to adversarial attacks object detection systems. It seems promising to combine YOLO architecture and diffusion-based models in this case.

5 Paper 1 Literature Review

5.1 Paper Title

YOLOv1 to YOLOv11: A Comprehensive Survey of Real-Time Object Detection Innovations and Challenges (2025)

5.2 Abstract/Paper Summary

This paper discusses the development process of YOLO models ranging from YOLOv1 to YOLOv11. The key improvements that have been made with regard to the architecture of YOLO, their application and future research prospects are outlined.

5.3 Introduction

The paper looks at the architectural evolution of YOLO models and compares the performance of these models to the benchmarks. The deployment of YOLO models on edge devices is discussed alongside some of the issues facing research in this area.

5.4 Dataset

The survey discusses three public benchmark datasets: MS COCO, PASCAL VOC, and ImageNet.

5.4.1 Dataset Links

Table 5: Public Benchmark Datasets Discussed in the Survey

Dataset	**Link**
MS COCO	`https://cocodataset.org/#download`
PASCAL VOC	`https://www.robots.ox.ac.uk/~vgg/projects/pascal/VOC/`
ImageNet	`https://www.image-net.org/`

5.5 Methodology

This research paper uses a survey approach in which various YOLO versions are compared in terms of their architectures, detection heads, training approaches, benchmarking performance (mAP and FPS), and practical implementations.

5.6 Future Direction / Limitation

Various challenges are outlined by the authors, such as robustness, training stability, model interpretability, and detecting smaller objects. Future research directions involve the development of even more effective YOLO models.

6 Paper 2 Literature Review

6.1 Paper Title

Hyper-YOLO: When Visual Object Detection Meets Hypergraph Computation (2025)

6.2 Abstract/Paper Summary

The current paper presents Hyper-YOLO, which is an advanced object detection model designed based on YOLO using hypergraphs for higher order relationship modeling between features. This model provides higher multi-scale feature fusion capabilities and achieves better detection accuracy compared to all available YOLO versions, particularly for smaller and more complicated objects.

6.3 Introduction

It is stated by the authors that there are certain drawbacks in feature fusion and cross-level interaction in traditional YOLO systems. For overcoming these drawbacks, the proposed Hyper-YOLO makes use of hypergraph computations in order to achieve better results.

6.4 Dataset

The proposed model is evaluated on the MS COCO public benchmark dataset.

6.4.1 Dataset Links

Table 6: Public Benchmark Dataset Used in Hyper-YOLO

Dataset	Link
MS COCO	`https://cocodataset.org/#download`

6.5 Methodology

The three main modules in the Hyper-YOLO algorithm are as follows: MANet acts as the backbone, HyperC2Net as the neck, and the HGC-SCS module acts as the hypergraph feature learning module. Together, these modules perform better than the previous YOLO algorithms in feature extraction and detection.

6.6 Future Direction / Limitation

Although Hyper-YOLO significantly improves detection accuracy, further research is needed to reduce computational complexity and optimize the model for efficient real-time deployment on resource-constrained devices.

7 Paper 3 Literature Review

7.1 Paper Title

Delving into YOLO Object Detection Models: Insights into Adversarial Robustness (2025)

7.2 Abstract/Paper Summary

This paper describes the evolution of YOLO architecture and analyzes its resilience against attacks as well as corrupted images. Comparison of different YOLO variants on the COCO dataset is made, and the most robust variants are selected.

7.3 Introduction

This research provides an overview of how YOLO models have evolved and explores the robustness of the YOLO models to adversarial attacks and noise in images. This is done by comparing several YOLO models on the COCO data set to find the most robust models for real-world object detection tasks.

It should be noted that while there is no lack of literature about YOLO models, few studies are devoted to the issue of robustness of YOLO models to adversarial attacks and noisy environments.

7.4 Dataset

The experiments are conducted using the MS COCO public benchmark dataset, which provides a fair comparison across different YOLO versions.

7.4.1 Dataset Links

Table 7: Public Benchmark Dataset Used in the Study

Dataset	Link
MS COCO	`https://cocodataset.org/#download`

7.5 Methodology

Evaluation of YOLOv3 is carried out using YOLOv11 framework against various attacks such as FGSM and PGD as well as image corruption methods. The performance of the models is measured using detection metrics such as IoU and AP.

7.6 Future Direction / Limitation

The authors recommend developing more robust YOLO models that can better resist adversarial attacks and image corruption, making them more reliable for safety-critical real-world applications.

8 Paper 4 Literature Review

8.1 Paper Title

YOLO Advances to Its Genesis: A Decadal and Comprehensive Review of the YOLO Series (2025)

8.2 Abstract/Paper Summary

This research paper presents an exhaustive discussion about the YOLO family of models, covering YOLOv1 to YOLOv12. The paper covers the changes made to architectures, improvements in performance, usage scenarios, and drawbacks of each YOLO version while discussing potential future directions for object detection systems.

8.3 Introduction

The authors highlight the widespread use of YOLO in computer vision and aim to provide a thorough analysis of YOLO's developments over the past decade. The paper provides an overview of the technology's development and its impact on current research in object detection.

8.4 Dataset

The paper reviews YOLO models evaluated on several public benchmark datasets, including PASCAL VOC, MS COCO, and ImageNet.

8.4.1 Dataset Links

Table 8: Public Benchmark Datasets Discussed in the Survey

Dataset	Link
MS COCO	`https://cocodataset.org/#download`
PASCAL VOC	`https://www.robots.ox.ac.uk/~vgg/projects/pascal/VOC/`
ImageNet	`https://www.image-net.org/`

8.5 Methodology

The analysis is conducted through the use of a chronological review approach that analyses the developments of YOLO architectures based on different aspects including backbones, accuracy, efficiency, and cross-domain uses.

8.6 Future Direction / Limitation

The authors suggest that future YOLO models should focus on multimodal learning, context-aware intelligence, improved computational efficiency, and integration with advanced AI systems to further enhance object detection performance.

9 Paper 5 Literature Review

9.1 Paper Title

ObjectAdv: Object-Level Unrestricted Adversarial Attacks via Diffusion Models (AAAI 2026)

9.2 Abstract/Paper Summary

In this research, we present ObjectAdv, which is an adversarial attack framework based on the diffusion process that focuses only on perturbation of the target object in the image without any modifications to the rest of the image. The proposed framework incorporates attention-based localization of objects, prompt switching, and edge smoothing mechanisms to generate realistic adversarial images with high success rates for attacks.

9.3 Introduction

Attack performance versus perceptual quality is one of the trade-offs observed in previous works on unrestricted adversarial attacks. To solve this problem, they have proposed a diffusion-based approach that only affects the semantic area of the object, hence creating more transferable and perceptually imperceptible adversarial examples.

9.4 Dataset

The proposed method is evaluated using the ImageNet-Compatible Dataset, which contains 1,000 validation images from ImageNet. The experiments are performed on multiple CNNs, Vision Transformers, and defense models.

9.4.1 Dataset Links

Table 9: Public Benchmark Dataset Used in ObjectAdv

Dataset	Official Link
ImageNet-Compatible Dataset (ImageNet Validation Set)	`https://www.image-net.org/`

9.5 Methodology

Objective Adv utilizes Stable Diffusion to generate unrestricted adversarial samples. The model is made up of four main building blocks: attention-based localization, prompt switching, edge smoothing using Fourier Transform, and a loss function optimization method that improves both efficiency and image quality.

9.6 Future Direction / Limitation

Despite the fact that the ObjectAdv model achieves high success rates of attacks and maintains high image quality, further research needs to be conducted to improve transferability of non-targeted attacks and robustness for different target models and defense methods.

10 Paper 6 Literature Review

10.1 Paper Title

DiffProtect: Generate Adversarial Examples with Diffusion Models for Facial Privacy Protection (2025)

10.2 Abstract/Paper Summary

In this research, we propose DiffProtect, an adversarial attack framework utilizing diffusion to produce natural images of faces in order to protect facial privacy. The approach leverages diffusion autoencoding along with semantic perturbation to generate natural images and significantly improve success rates compared to current state-of-the-art face encryption approaches.

10.3 Introduction

According to the authors, existing facial privacy techniques using noise, masking, and GAN methods have shortcomings, as they reduce the quality of images or provide insufficient attack success. Therefore, the new model uses diffusion models to create semantically plausible adversarial attacks that achieve high success rates while keeping high quality..

10.4 Dataset

The proposed framework is evaluated on two public facial image datasets: CelebA-HQ and FFHQ, using 1,000 images from each dataset.

10.4.1 Dataset Links

Table 10: Public Datasets Used in DiffProtect

Dataset	**Official Link**
CelebA-HQ	`https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html`
FFHQ (Flickr-Faces-HQ)	`https://github.com/NVlabs/ffhq-dataset`

10.5 Methodology

DiffProtect uses the combination of Diffusion Autoencoder (DiffAE) and a conditional DDIM decoder for optimization of adversarial semantic codes. Moreover, DiffProtect introduces the concept of face semantics regularization for maintaining facial identity, and DiffProtect-fast for faster processing with reduced computation overheads and similar attack effectiveness.

10.6 Future Direction / Limitation

While having obtained high success rates for attacks and generating images that have natural look, the initial approach is limited by poor computational performance due to its iterative nature. Further research should focus on improving the computational efficiency without compromising image quality and effectiveness of attacks.

11 Paper 7 Literature Review

11.1 Paper Title

DiffAdvMAP: Flexible Diffusion-Based Framework for Generating Natural Unrestricted Adversarial Examples (ICML 2025)

11.2 Abstract/Paper Summary

In this work, DiffAdvMAP is proposed, which is a novel diffusion-based method for constructing unconstrained adversarial examples (UAEs). This method uses MAP optimization in the latent space of the diffusion process to create adversarial images that have high success rates, excellent transferability properties, and increased robustness against defenses.

11.3 Introduction

The authors note that the present unrestricted attacks usually face challenges of being effective, image quality, and efficient all at once. To solve such problems, DiffAdvMAP utilizes diffusion models along with MAP optimization in order to generate realistic adversarial examples that can reach high transferability.

11.4 Dataset

The experiments are conducted on an ImageNet-compatible dataset, consisting of 1,000 randomly selected images (one image per ImageNet class) for evaluating unrestricted adversarial attacks.

11.4.1 Dataset Links

Table 11: Public Dataset Used in DiffAdvMAP

Dataset	Official Link
ImageNet	`https://www.image-net.org/`

11.5 Methodology

DiffAdvMAP integrates diffusion models and MAP optimization to sample latent codes with adversarial properties from the posterior distribution. It uses adversarial regularization, reconstruction regularizations, and a destruction-reconstruction technique to improve the efficiency and effectiveness of attacks on images.

11.6 Future Direction / Limitation

Despite DiffAdvMAP achieving effective 100% white-box attack success rate along with maintaining the natural look of images, future research efforts must concentrate on black-box transferability improvement, optimization of the computational process, and increase in robustness to state-of-the-art defense techniques.

12 Paper 8 Literature Review

12.1 Paper Title

A Study of YOLO, Transformer and Diffusion Model for Small Object Detection (2025)

12.2 Abstract/Paper Summary

This paper will give an elaborate analysis of the three major deep learning frameworks that include YOLO, Transformer, and Diffusion model for detecting small objects. The comparison will be done on the basis of the architecture, detection abilities, strengths, weaknesses, and future research avenues.

12.3 Introduction

The authors argue that there are still difficulties with small object detection due to the few number of pixels available and also due to the complexity of backgrounds. The paper presents a comprehensive analysis of YOLO, Transformers, and diffusion techniques in order to understand their advantages, disadvantages, and future possibilities of development in small object detection.

12.4 Dataset

The review discusses experiments conducted on several public benchmark datasets, including MS COCO, PASCAL VOC, CrowdHuman, VisDrone, and their diffusion-extended versions (COCO-DE and VOC-DE).

12.4.1 Dataset Links

Table 12: Public Datasets Discussed in the Survey

Dataset	**Official Link**
MS COCO	`https://cocodataset.org/#download`
PASCAL VOC	`https://www.robots.ox.ac.uk/~vgg/projects/pascal/VOC/`
CrowdHuman	`https://www.crowdhuman.org/`
VisDrone	`https://aiskyeye.com/`
COCO-DE / VOC-DE	Generated from MS COCO and PASCAL VOC using diffusion-based data generation.

12.5 Methodology

In this study, the following three techniques will be reviewed as representatives of the approaches used for detecting small objects: YOLO (focusing on multi-scale feature fusion and optimization of the network structure), Transformer (using global attention mechanism and hybrid models), and diffusion models (such as DiffusionDet, DiffCOD, and DiffusionEngine).

12.6 Future Direction / Limitation

Diffusion models have higher detection precision but suffer from high computational expenses, time consumption, and difficult learning process, as mentioned by the authors. Future work needs to focus on the light-weighted model architecture, multi-modal feature integration, effective real-time application, and low dependency on data while maintaining high detection efficiency.

13 Conclusion

A synthesis of eight papers of the latest advances (2025–2026) in the field of object detection via YOLO and diffusion-based attacks generation will be provided in this literature review. As is seen from the YOLO papers review, because of the continuous improvements in the sphere of feature fusion, hypergraph computation, and detection, considerable progress has been achieved in the area of accuracy, computational efficiency, and real-time object detection. It should be pointed out that, even taking into account all the advantages, latest YOLO models are not resistant to adversarial attacks.

From the review of diffusion models, it may be noted that diffusion models prove to be quite efficient when it comes to the generation of highly visual realistic adversarial examples with the preservation of high quality. The recent models like ObjectAdv, DiffProtect, and DiffAdvMAP produce highly successful attacks compared to previous methods. Besides, there exists a great potential of diffusion models in the development of small object detection due to their good feature generation and data augmentation capabilities. Unfortunately, the high computational cost and low inference speed constitute serious obstacles for the implementation of this method.

In conclusion, it may be stated that further researches in the sphere of object detection should be aimed at the development of efficient, accurate, light, and resistant to adversarial attacks object detection systems. It seems promising to combine YOLO architecture and diffusion-based models in this case.